



University of Wisconsin
SCHOOL OF MEDICINE
AND PUBLIC HEALTH

Poisson Flow Model of Cortical Folding Pattern

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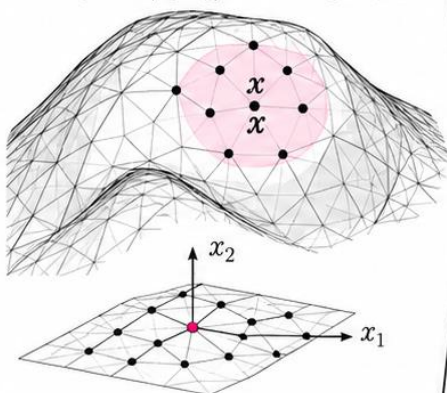
<https://github.com/laplcebeltrami/poisson>

Mean Curvature on the Cortical Surface

Local quadratic fitting $\rightarrow$ fundamental forms $\rightarrow$ mean curvature $\rightarrow$ meah-centering

1 Local neighborhood and coordinates

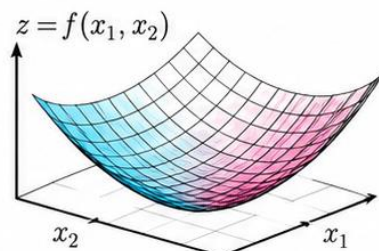
At each vertex x , take the first-ring neighborhood and represent the surface in a local 2D coordinate system (x_1, x_2) on the tangent plane.



2 Quadratic polynomial fit

Fit a quadratic surface to the neighboring vertices by least squares:

$$f(x_1, x_2) = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \frac{1}{2} \beta_3 x_1^2 + \beta_4 x_1 x_2 + \frac{1}{2} \beta_5 x_2^2.$$



Estimate coefficients $(\beta_0, \dots, \beta_5)$ from the neighboring vertices.

3 First and second fundamental forms

From the fitted surface, compute the first and second fundamental forms:

$$g = \begin{pmatrix} 1 + \beta_1^2 & \beta_1 \beta_2 \\ \beta_1 \beta_2 & 1 + \beta_2^2 \end{pmatrix},$$

$$h = \begin{pmatrix} \beta_3 & \beta_4 \\ \beta_4 & \beta_5 \end{pmatrix}.$$

Mean curvature at vertex x :

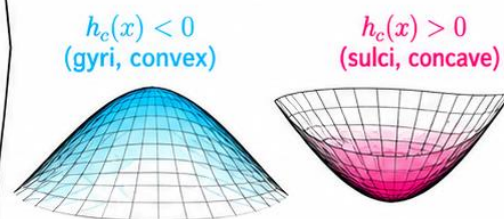
$$h(x) = \frac{1}{2} \text{tr}(g^{-1}h).$$

4 Mean-centering

Subtract the area-weighted mean to focus on spatial variations:

$$h_c(x) = h(x) - \frac{1}{\text{Area}(\mathcal{M})} \int_{\mathcal{M}} h(x) d\mu(x),$$

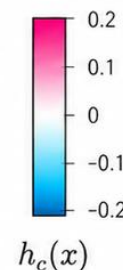
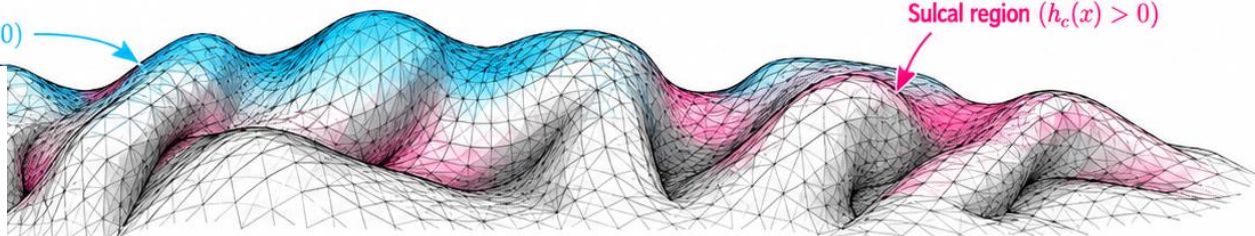
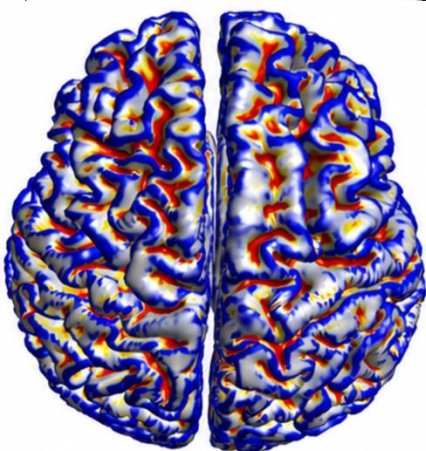
so that $\int_{\mathcal{M}} h_c(x) d\mu(x) = 0.$



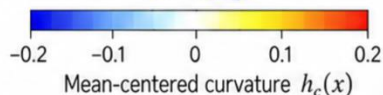
Example: mean-centered curvature on a cortical surface

Gyral region ($h_c(x) < 0$)

Sulcal region ($h_c(x) > 0$)

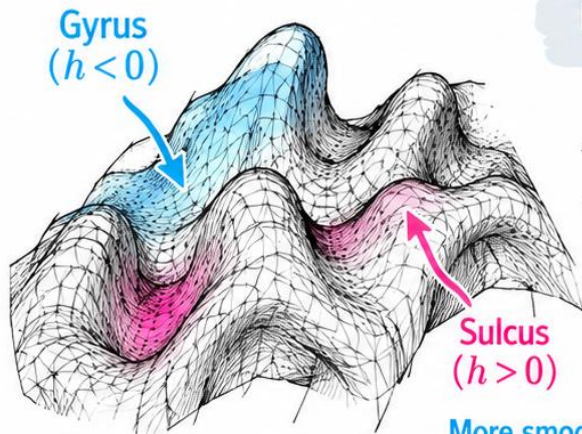


Curvature estimation is noisy.
How to reduce noise?



Poisson Model of Cortical Folding

Screened Poisson smooths the curvature-driven pattern



Mean curvature $h(x)$

Poisson equation
($\lambda = 0$)

$$\mathcal{L}u = h \quad \text{on } \mathcal{M}$$

- Smoothing potential for curvature field h
- Solution defined up to a constant

Screened Poisson equation
($\lambda > 0$)

$$\mathcal{L}u + \lambda u = h \quad \text{on } \mathcal{M}$$

- Adds a stabilizing term
- Unique solution
- Controls the amount of smoothing

As $\lambda \rightarrow 0$

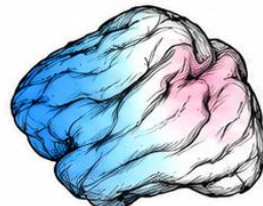
- recovers the Poisson equation
- stronger smoothing (low-frequency dominance)

As λ increases

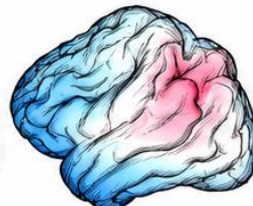
- less smoothing
- retains more fine-scale folding patterns

More smoothing
(low-frequency)

Less smoothing
(higher-frequency) $u(x)$



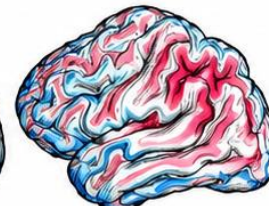
$\lambda = 0$
(Poisson)



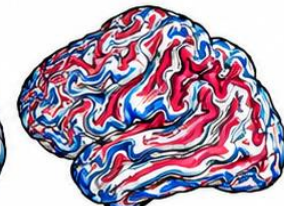
$\lambda = 0.01$



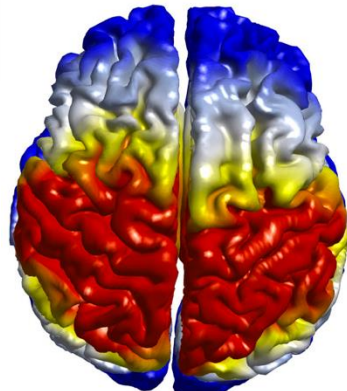
$\lambda = 0.1$



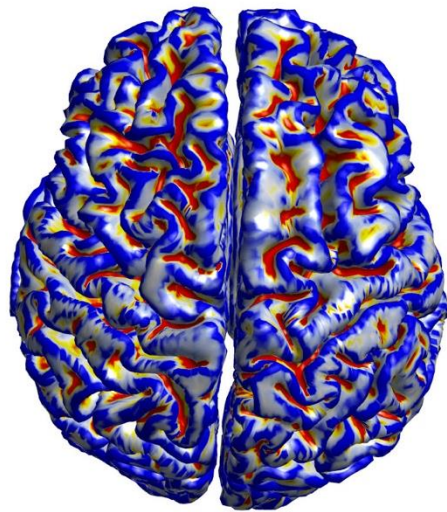
$\lambda = 1$



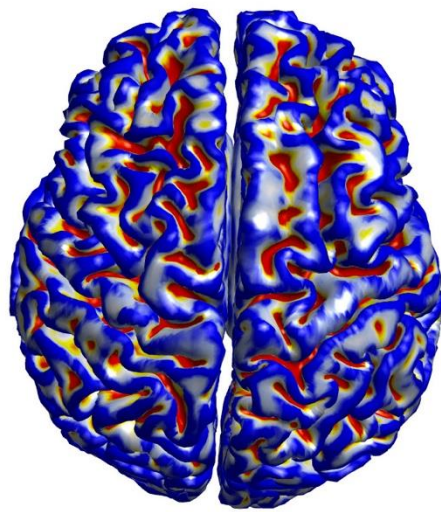
$\lambda = 10$



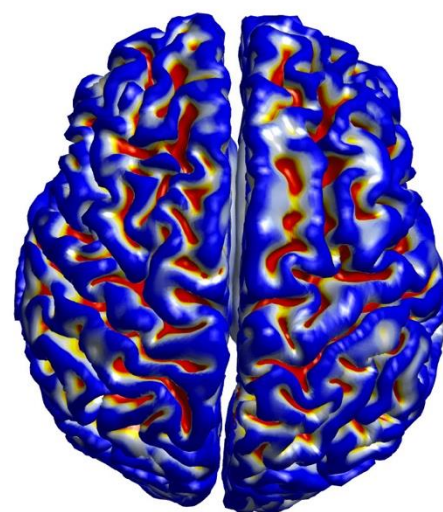
Screened Poisson Equation



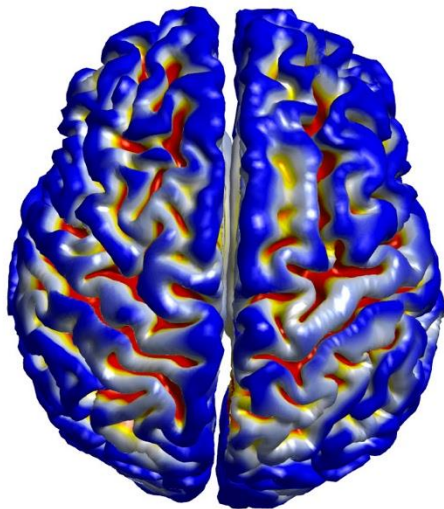
Original



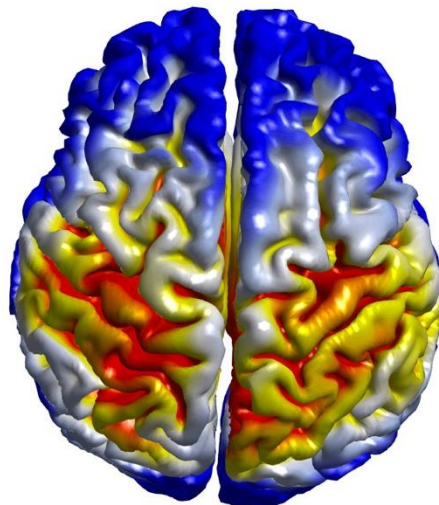
1.0



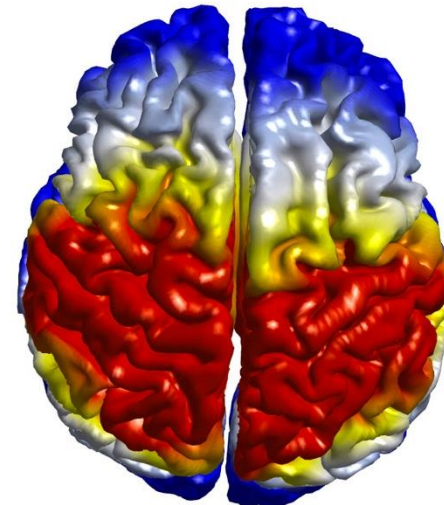
0.1



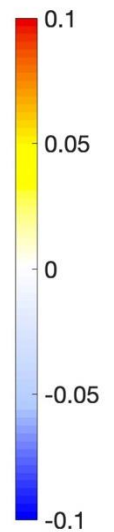
0.01



0.001



0

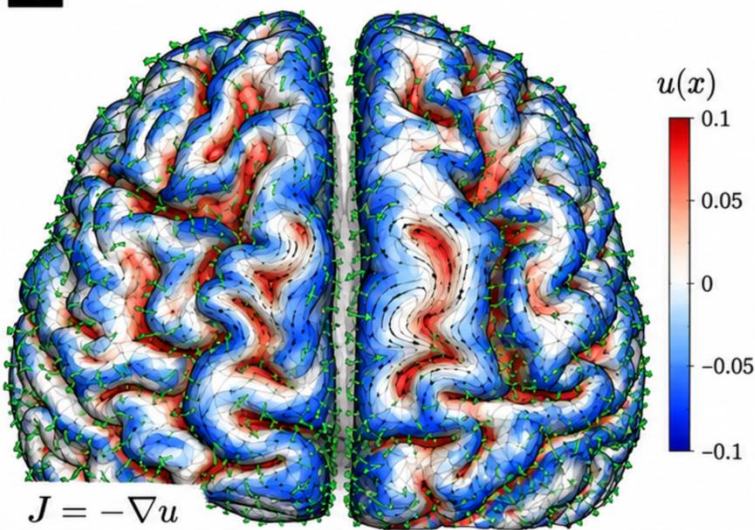


Flux field of screened Poisson equation ($\lambda = 0.1$)

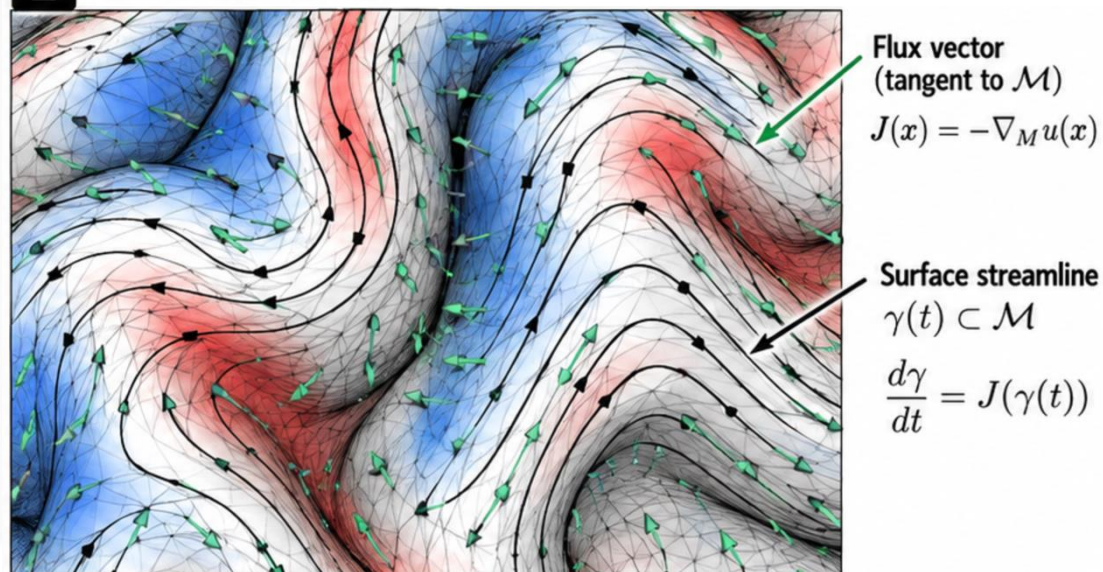
$$(\mathcal{L} + \lambda I)u = h$$

$$J = -\nabla u$$

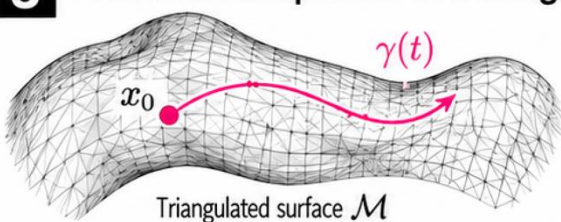
1 Flux vector field on cortical surface



2 Flux streamlines



3 Streamline computation on a triangulaal mesh



Streamline equation on $\mathcal{M}$

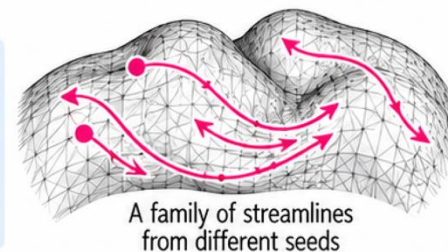
$$\frac{d\gamma}{dt} = J(\gamma(t)), \quad \gamma(t) \subset \mathcal{M}$$

$$J = -\nabla u \quad (\text{tangent vector field})$$

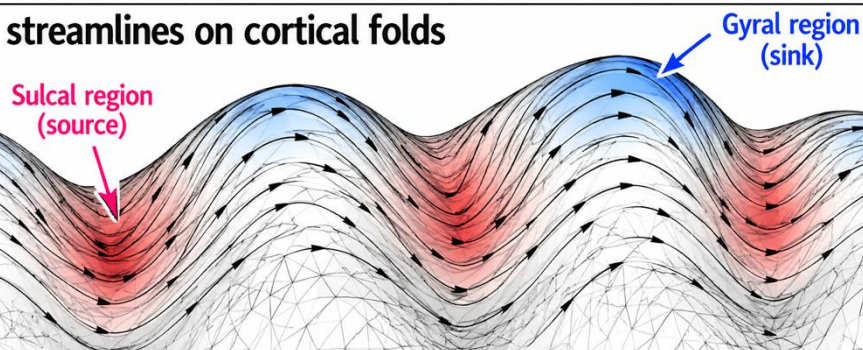
Discrete integration (e.g., Runge-Kutta)

$$\gamma_{k+1} = \gamma_k + \delta t \cdot J(\gamma_k)$$

δt : step size



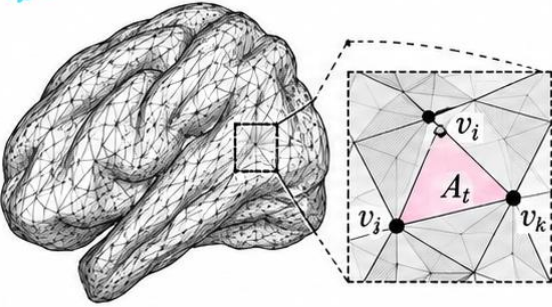
Surface streamlines on cortical folds



Numerical Implementation

Solve the screened Poisson equation on a triangulated cortical surface

1 Triangulated surface and FEM matrices

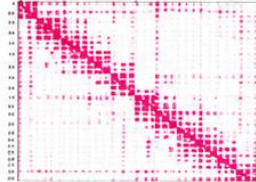
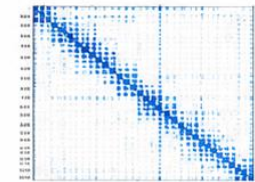


Triangulated cortical mesh $\mathcal{M}$

Finite-element matrices (cotangent discretization)

Mass matrix A

Stiffness matrix C



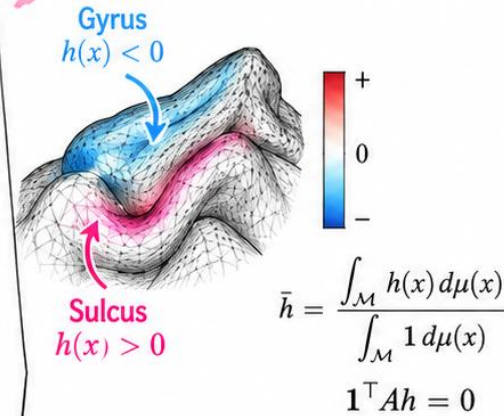
$$\mathbf{1}^\top A f = \int_{\mathcal{M}} f(x) d\mu(x)$$

(surface integral)

$$u^\top C v = \int_{\mathcal{M}} \nabla u \cdot \nabla v d\mu(x)$$

(Dirichlet energy)

2 Mean-center the curvature field

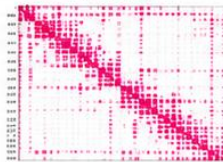


3 Discrete screened Poisson equation

$$(C + \lambda A)u = Ah \quad (8)$$

$\lambda > 0$: regularization parameter

$C + \lambda A$



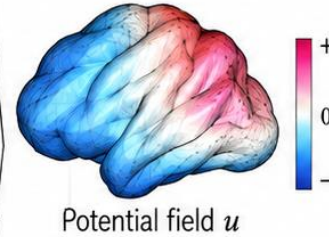
Solve using sparse
Cholesky factorization

$\longrightarrow u$

4 Normalize the solution

$$u \leftarrow \frac{\|h\|}{\|u\|} u$$

$$\|f\|^2 = \frac{1}{\text{Area}(\mathcal{M})} f^\top A f \quad (9)$$

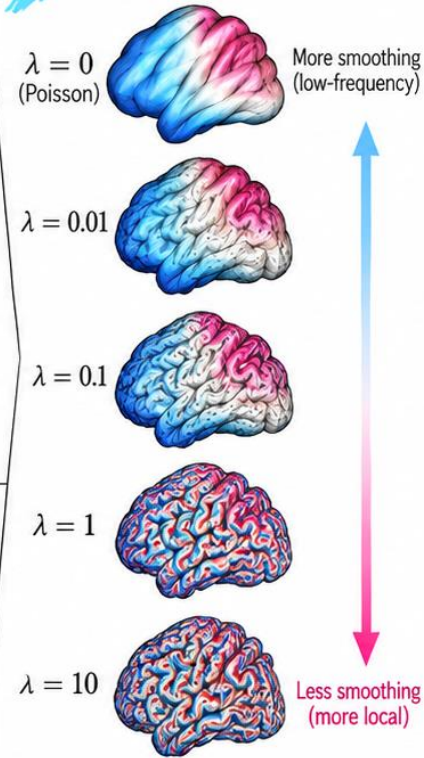


5 Zero-mean condition

$$\mathbf{1}^\top A u = 0 \quad (10)$$

Removes the additive constant
and provides a consistent
reference level across subjects.

6 Effect of λ (spatial scale)



Heat Diffusion vs. Screened Poisson Filtering

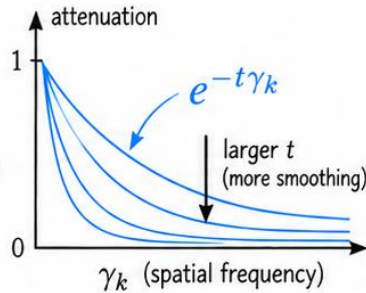
Same goal: smoothing. Different spectral attenuation.

1 Heat diffusion

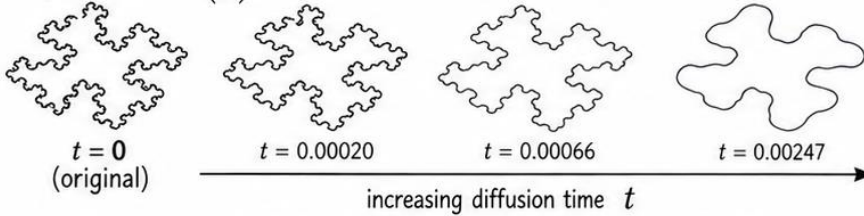
$$\frac{\partial f}{\partial t} = -\mathcal{L}f, \quad f(x, 0) = h(x)$$

$$f(x, t) = e^{-t\mathcal{L}}h(x) = \sum_{k=0}^{\infty} e^{-t\gamma_k} \langle h, \phi_k \rangle \phi_k(x)$$

Spectral response: $e^{-t\gamma_k}$



Original curve $h(x)$

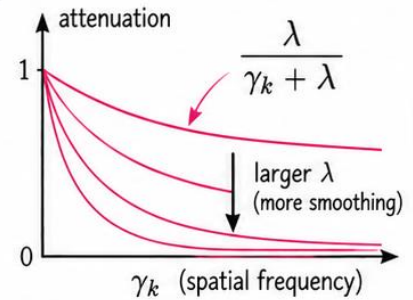


2 Screened Poisson equation

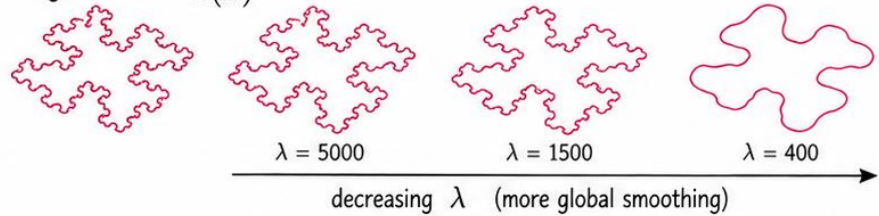
$$(\mathcal{L} + \lambda I)u = h$$

$$u(x) = \sum_{k=0}^{\infty} \frac{1}{\gamma_k + \lambda} \langle h, \phi_k \rangle \phi_k(x)$$

Spectral response: $\frac{\lambda}{\gamma_k + \lambda}$



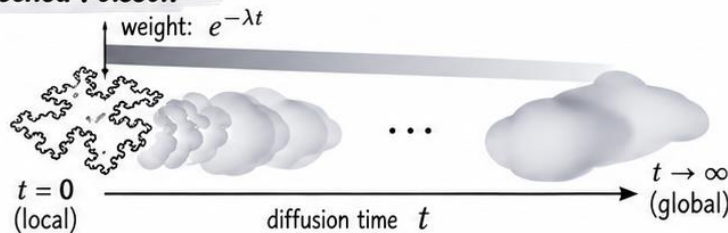
Original curve $h(x)$



3 Heat-kernel representation of screened Poisson

$$(\mathcal{L} + \lambda I)^{-1} = \int_0^{\infty} e^{-\lambda t} e^{-t\mathcal{L}} dt.$$

Screened Poisson = exponentially weighted superposition of heat diffusion over all times.



4 Low-frequency matching

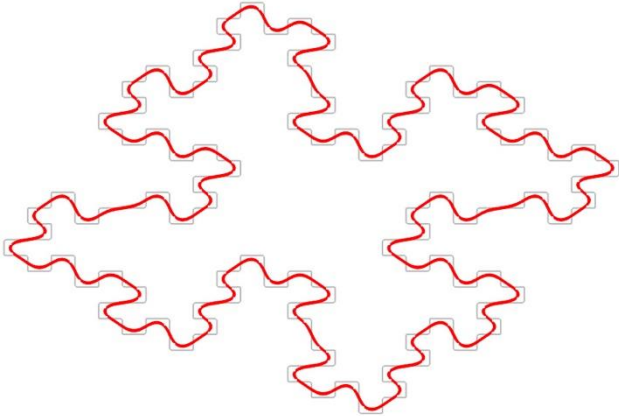
$$e^{-t\gamma_k} = \frac{\lambda}{\gamma_k + \lambda} \quad (\gamma_k \approx 0)$$

$$e^{-t\gamma_k} = 1 - \frac{\gamma_k}{\lambda} + O(\gamma_k^2),$$

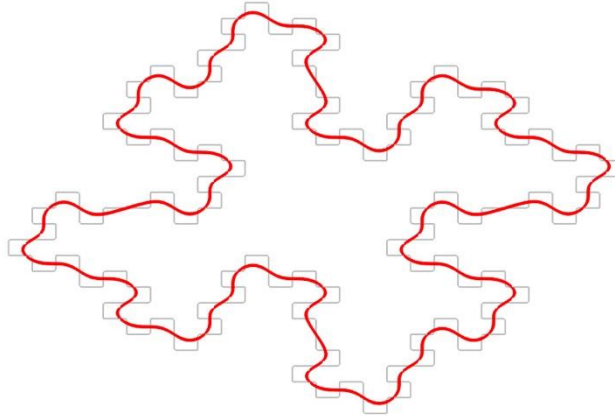
$$\frac{\lambda}{\gamma_k + \lambda} = 1 - \frac{\gamma_k}{\lambda} + O(\gamma_k^2).$$

Spectral attenuation in heat diffusion & screened Poisson

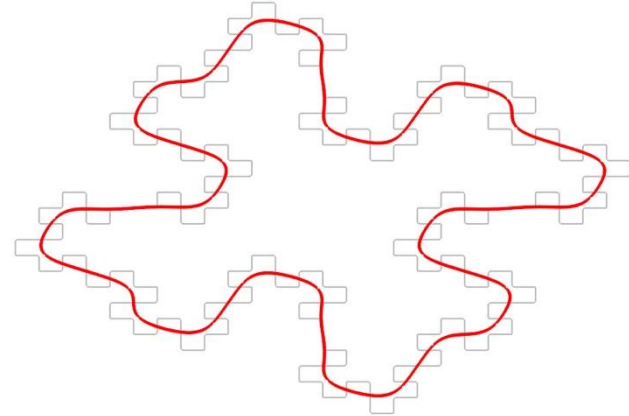
Heat: $t = 0.00020$



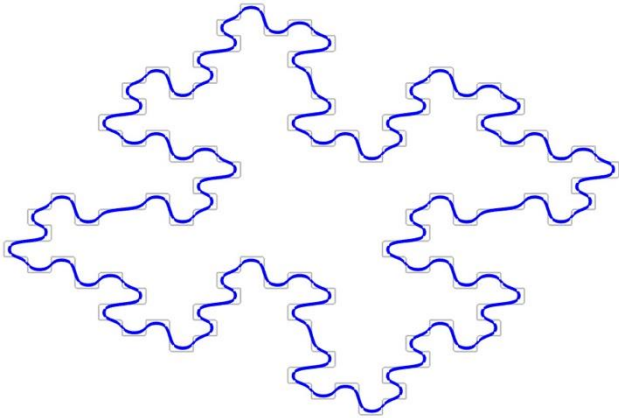
Heat: $t = 0.00066$



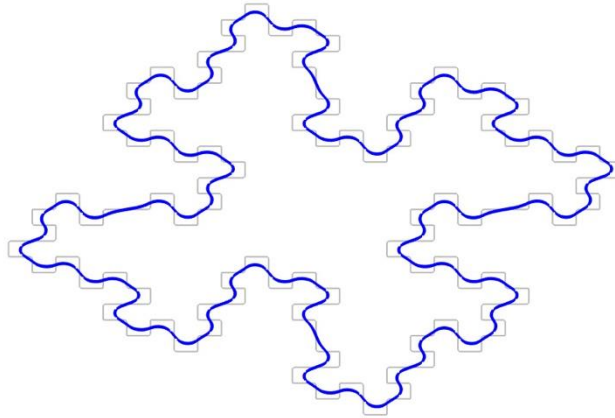
Heat: $t = 0.00247$



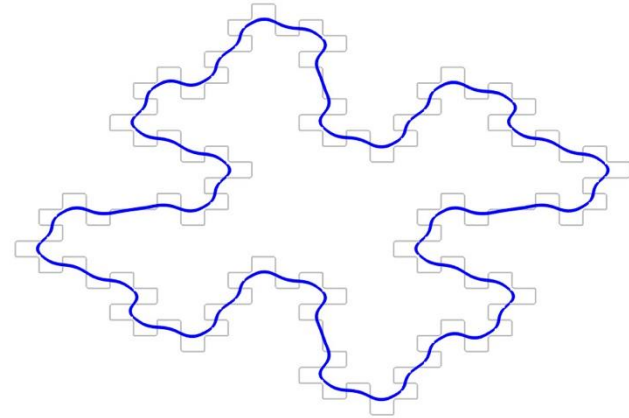
Poisson: $\lambda = 5000$



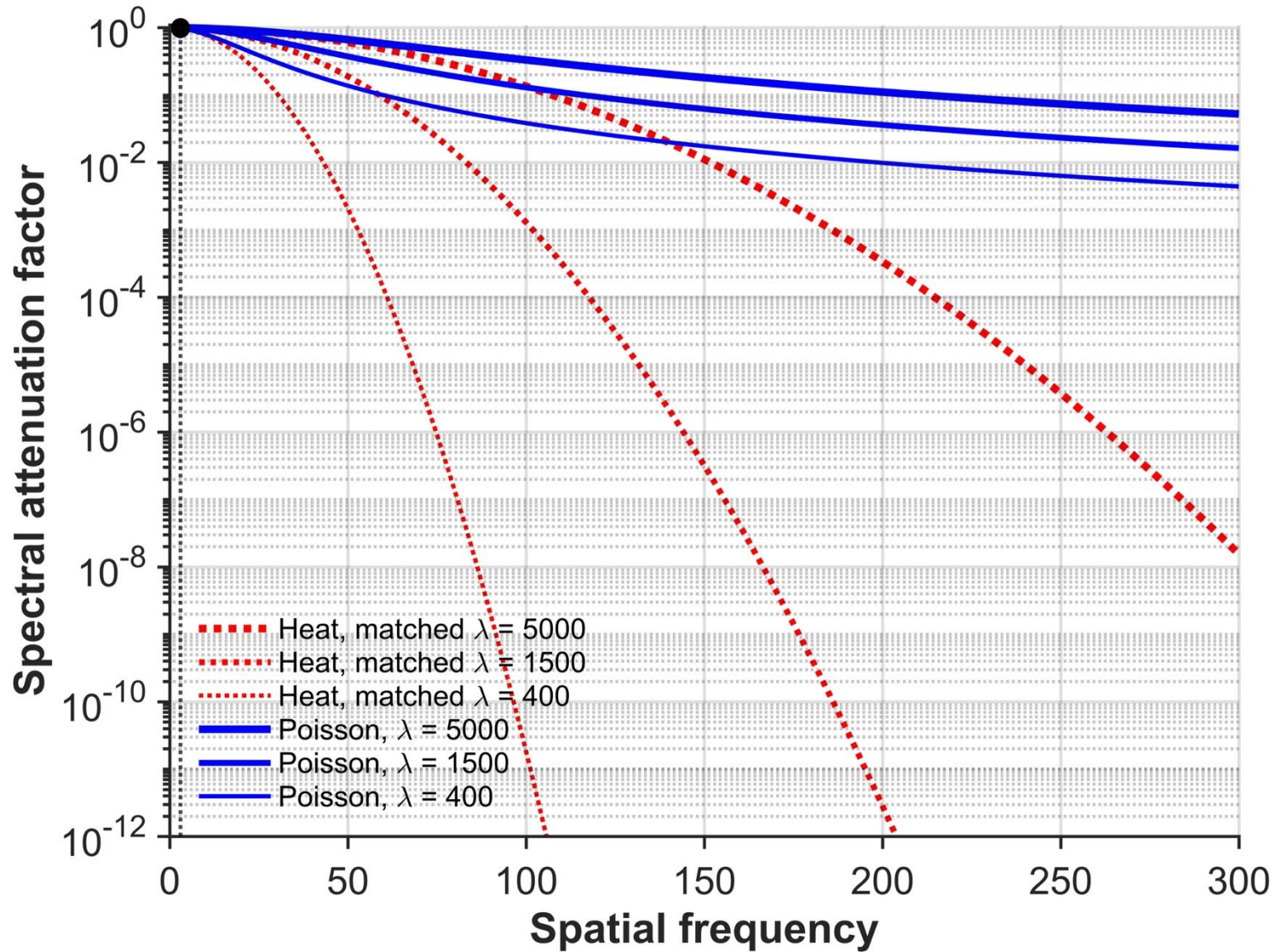
Poisson: $\lambda = 1500$



Poisson: $\lambda = 400$

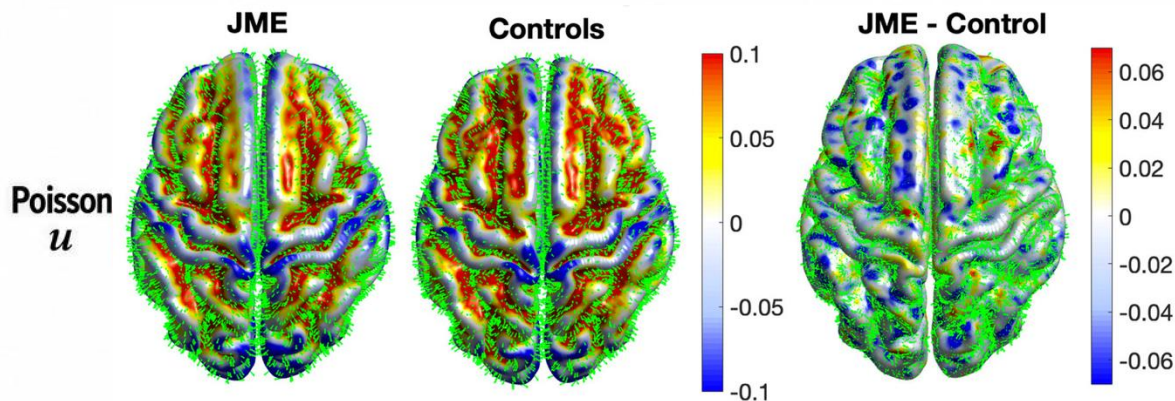


Spectral attenuation in heat diffusion & screened Poisson



Group level analysis on surface

1 Group mean of Poisson potential



2 Vertexwise general linear model

$$u = c_0 + c_1 \text{ Group} + c_2 \text{ Age} + c_3 \text{ Sex} + \varepsilon \quad (11)$$

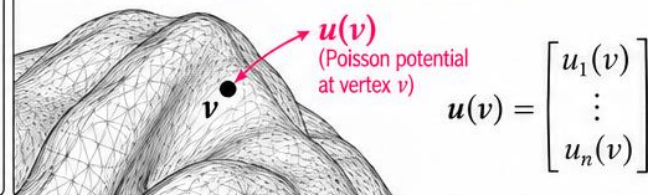
$u(v)$: Poisson potential at vertex v

Group : JME vs. control

Age, Sex : nuisance covariates

Test group effect using vertexwise t -statistic

$$t(v) = \frac{\hat{c}_1(v)}{\text{SE}(\hat{c}_1(v))}$$



3 GLM results accounting for covariates

JME - Controls
(Age, Sex)



JME - Controls
(Age only)



Left lateral

Left medial

Dorsal

Right medial

Right lateral

Poisson:
JME-Controls |
Age, Sex

